

Quiz #17

You need to use the ideal gas law to calculate temperature for various gases. The gas constant R changes depending on the gas being analyzed. Ask the user to input the gas being tested. Use the function for the ideal gas law (shown below) and string concepts to allow for the user-defined gas to be used when calling the function. The pressure, density and R values are found below.

The following is an example of what the input prompt should say:

Enter the chemical name for the gas (Ar,CO₂,CO,H₂,N₂,O₂,Air):

The following output is an example of what the output should look like:

The temperature in Kelvins is 397.5768 K for the gas Ar.

Needed information:

pressure= 101205 N/m²

density=1.22382 kg/m³

Gas	R
Argon, Ar	208
Carbon Dioxide, CO ₂	188.9
Carbon Monoxide, CO	297
Hydrogen, H ₂	4124
Nitrogen, N ₂	296.8
Oxygen, O ₂	259.8
Air	286.9

```
function [ t ] = ideal_gas_law2( p,R,rho )
%use ideal gas law to find temperature at a point
t=p/(rho*R);
end
```

